

EMS Stroke Tip Sheet – AIS Recognition & Transfer Planning

Note: Specific procedures and interventions may vary based on EMS agency protocols and must be performed in accordance with guidance from the service's medical director.

Always follow your approved local protocols and scope of practice.

Initial Stroke Recognition and Assessment

- Perform validated stroke scale: NIHSS, otherwise VAN for LVO detection.
- If VAN positive, consider patient high-risk for Large Vessel Occlusion (LVO):
 - Recommend immediate CT Head AND CT Angiography of Head & Neck
 - Notify nearest Comprehensive Stroke Center with Neuro-Interventional availability
- If VAN positive and comprehensive center is within 15 minutes, proceed directly; otherwise, go to nearest Acute Stroke Ready Hospital.

Information to Communicate to ED/Stroke Team

- Time of Last Known Normal (LKW)
- VAN result (+/-) and/or NIHSS if available
- Complete medication list (esp. anticoagulants)
- Pertinent PMH: prior stroke, recent surgery, bleeding disorders, anticoagulant use
- Blood glucose and vital signs

Thrombolytic Screening – TNK/Alteplase Candidate?

- Confirm inclusion/exclusion criteria:
 - LKW <4.5 hours
 - Head CT negative for hemorrhage
 - No anticoagulants or recent bleeding
 - BG not <50 or >400 mg/dL
 - BP <185/110 mmHg (must be lowered in order to receive thrombolytic therapy)

EMS Prehospital Checklist – AIS

- Establish 2 sites of IV access if possible (one dedicated for thrombolytic)
- Monitor blood pressure closely
- Keep patient NPO
- Continuous cardiac and pulse oximetry monitoring

Interfacility Transport – “Drip and Ship” Considerations

For patients who received thrombolytics (Alteplase or Tenecteplase) at an outside hospital and are being transferred for higher-level care (e.g., thrombectomy or ICU):

◇ Medication Type Differences:

- **Alteplase (tPA):** Administered as a 10% bolus + 90% continuous infusion over 60 minutes. EMS must monitor the infusion pump if still running during transport.

- **Tenecteplase (TNK):** Given as a single IV bolus over 5 seconds. No ongoing infusion needed during transport.

◇ Monitoring and Vital Signs:

- Neuro checks and vital signs every 15 minutes during transfer.
- Maintain **SBP <180 mmHg** and **DBP <105 mmHg**.
- Watch for acute neurological decline, nausea, vomiting, severe headache, or signs of bleeding.
- If signs of deterioration occur, contact medication director to consider stopping alteplase infusion (if applicable), notify receiving facility, and prepare for possible management of intracranial hemorrhage.

◇ Angioedema:

- A rare but serious adverse effect, usually within the first 2 hours post-thrombolytic.
- Signs: swelling of lips, tongue, or airway. May affect only one side.
- Management:
 - Discontinue thrombolytic if infusing
 - Administer antihistamines, corticosteroids, epinephrine if needed
 - Secure airway early if symptoms progress

◇ Antihypertensive Management (if BP > 180/105 mmHg):

- Preferred agents:
 - **Labetalol** 10–20 mg IV over 1–2 minutes, may repeat or double dose
 - **Nicardipine** IV infusion starting at 5 mg/hr
- Avoid hypotension (SBP <110 mmHg)

◇ Blood Thinners – What Counts?

- **Anticoagulants (contraindicate thrombolytics):**
 - Warfarin (Coumadin)
 - Apixaban (Eliquis)

- Rivaroxaban (Xarelto)
 - Dabigatran (Pradaxa)
 - Edoxaban (Savaysa), Betrixaban (Bevyxxa)
 - Enoxaparin (Lovenox), Fondaparinux (Arixtra), Heparin (UFH)
 - ****Antiplatelets (not a contraindication alone):****
 - Aspirin, Clopidogrel (Plavix), Ticagrelor (Brilinta), Prasugrel (Effient)
 - Dipyridamole/Aspirin (Aggrenox), Cangrelor (IV), Ticlopidine (rare)
- ◇ Key Reminders:
- Continue NPO status during transport.
 - Maintain two good IV lines if possible.
 - Ensure proper handoff and documentation on arrival to receiving facility.